

BIOS and POST Error Troubleshooting

Assignment Answers

1. Five common BIOS or POST error messages

Error message	Usual meaning / likely issue
CMOS checksum error – Defaults loaded	Stored BIOS/CMOS settings failed validation; a weak CMOS battery or corrupted settings may be responsible.
Keyboard error or No keyboard detected	Keyboard is disconnected, faulty, connected to a problematic port, or not detected by firmware.
CPU fan error	CPU fan is unplugged, failed, running too slowly, or connected to the wrong motherboard header.
No bootable device / Operating system not found	Boot drive is missing, disconnected, failed, or the boot order/UEFI mode is incorrect.
Memory error / RAM not detected	RAM module may be loose, incompatible, defective, or the motherboard memory slot may have a problem.

2. Three repeating beeps, no display: AMI BIOS

In classic AMI BIOS beep-code tables, **three short beeps typically indicate a base-memory (first 64 KB) read/write test failure**, pointing first toward system RAM or its connection. Beep-code meanings can vary by motherboard model and BIOS/UEFI generation, so check the exact PC or motherboard manual before treating this as definitive. [1][2]

Troubleshooting step 1: Turn the PC off, unplug it, and discharge residual power by holding the power button briefly. Follow ESD precautions, then remove and firmly reseat the RAM modules. Check that each module's latches lock into place.

Troubleshooting step 2: Test one RAM module at a time in the motherboard-recommended slot. If possible, try a known-compatible working module. If the beeps continue, inspect the memory slots and consult the board manual; the motherboard or memory controller may also be involved.

References: [1] ASRock, "What's the meaning of AMI BIOS Beep Codes?" <https://www.asrock.com/support/faq.asp?k=beep+codes> | [2] Intel, "Computer Beeps 3 Times on Startup for Intel Desktop Boards" <https://www.intel.com/content/www/us/en/support/articles/000005765/boards-and-kits/desktop-boards.html>

3. Entering BIOS setup and identifying manufacturer/version

I cannot access a physical PC or virtual machine in your environment, so the BIOS screen below is an **illustrative example only**, not a real screenshot from your computer. For your practical, replace the example with a photo/screenshot of your own BIOS Main/System Information page.

UEFI BIOS Setup Utility (ILLUSTRATIVE SAMPLE)

Main

System Date [09/30/2026]
System Time [20:00:00]
Motherboard Example Manufacturer (replace with actual)
BIOS Version Example Version 1.00 (replace with actual)
Processor Example CPU
Memory 16384 MB

Sample screen for layout guidance only — values are not actual hardware information.

How to capture your own screen: Restart the PC and tap the setup key shown during startup (commonly Del, F2, Esc, or F10; it varies by manufacturer). Open Main, Information, or System Summary. Record the motherboard/system manufacturer and BIOS/UEFI version, then take a photo or use the VM's screenshot function. Some systems show the PC manufacturer rather than the motherboard maker.

4. “CMOS checksum error – Defaults loaded”

The BIOS checks stored configuration data when the computer starts. This message means the stored CMOS/BIOS settings did not pass the checksum check, so the firmware loaded default settings. A weak or depleted CMOS coin-cell battery is a common cause, especially if the date and time reset; corrupted settings or an interrupted BIOS update can also cause it. [3][4]

- Enter BIOS/UEFI Setup and choose the option to load optimized/default settings, then save and restart.
- Set the correct date and time, and verify settings such as boot order, storage mode, and UEFI/Legacy configuration if applicable.
- If the error returns or the clock resets after the PC is unplugged, shut down and disconnect power. Replace the CMOS battery with the exact type specified by the motherboard/system manual (commonly CR2032 on desktops), observing polarity and ESD precautions.
- Start the PC, enter BIOS again, set date/time and required settings, save, and restart. If the warning persists with a new battery, consult the manufacturer; firmware corruption or a motherboard fault may require service.

References: [3] HP Support, “Resolve BIOS ROM or CMOS checksum error” https://support.hp.com/ph-en/document/ish_9432008-9432063-16 | [4] BIOSTAR Support FAQ, “What is BIOS screen error message means?” <https://www.biostar-usa.com/app/en-us/support/faq.php?page=21&searchfaq;=ati>

Safety note: Power off and unplug a desktop before reseating RAM or replacing a CMOS battery. Follow the system manual, use ESD precautions, and never force a component into its slot.